

Web Development Full Study Note - 2025

1. Web Evolution: Web 1.0 to 4.0

Web 1.0: Static, read-only pages.

Web 2.0: Interactive, social media, user content.

Web 3.0: Semantic web, personalized, AI-powered.

Web 4.0: Intelligent, IoT-connected, real-time.

2. Internet Protocols

TCP: Reliable data delivery.

DNS: Translates domain names to IP.

FTP: Transfers files between computers.

3. HTTP vs HTTPS

HTTP: No encryption, not secure.

HTTPS: Encrypted with SSL/TLS, secure for sensitive data.

4. Dynamic Content Delivery

Generated in real time using server-side code and databases.

5. Web Languages

HTML: Structure.

CSS: Styling.

JavaScript: Interactivity.

PHP: Server-side scripting.

6. Web Development Architecture

3-Tier: Presentation (frontend), Logic (backend), Data (database).

Useful for organizing code and scalability.

7. Common Development Issues

Web Development Full Study Note - 2025

Compatibility, performance, responsiveness, SEO, security flaws.

8. Scripting Languages

Client-side: JavaScript.

Server-side: PHP.

Shell: Bash.

9. Types of CSS

Inline: In tag.

Internal: In <style>.

External: In .css file.

10. JavaScript Basics

Integer, Double, Variable, String, Boolean examples explained.

11. JavaScript Operators

Arithmetic, Assignment, Comparison, Logical, String.

12. Web Hosting & Domains

Domain: Website address.

Hosting: Shared, VPS, Dedicated, Cloud.

13. Virtualization & Containers

VM: Full OS in virtual environment.

Container: Lightweight app packaging (Docker).

14. Application Security

SQLi: Attack via SQL inputs.

XSS: Script injection.

Web Development Full Study Note - 2025

DoS: Server overload attacks.

15. Cryptography & HTTPS

Encryption, TLS/HTTPS for secure web communication.

Hashing for password storage.

16. API Design - RESTful APIs

GET, POST, PUT, PATCH, DELETE methods.

Stateless access to resources over HTTP.